Subspace Inference for Bayesian Deep Learning

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Problem: Bayesian Inference in Deep Learning

- Bayesian methods provide accurate predictive distributions and well-calibrated uncertainty
- But modern DNNs have extremely high-dimensional parameter spaces ($\sim 10^6 - 10^9$)
- Standard Bayesian inference (MCMC, VI) doesn't scale well
- **Challenge:** How to perform efficient Bayesian inference in high-dimensional spaces?

Key Idea: Subspace Inference (SI)

Instead of working in the full parameter space $\mathbb{R}^d$:

Subspace parameterization

$$w = \hat{w} + Pz, \quad z \in \mathbb{R}^K, \quad K \ll d$$

where:

- $\hat{w}$: shift vector
- P: projection matrix defining the subspace
- z: low-dimensional parameters for inference

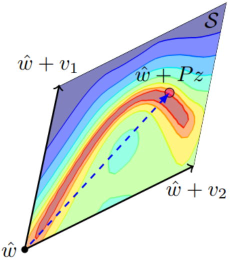


Figure: Illustration of subspace $\mathcal{S}$ with shift vector $\hat{w}$ and basis vectors v_1, v_2

Bayesian Model Averaging in the Subspace

$$p(D \mid z) = p_M(D \mid w = \hat{w} + Pz)$$

$$p(D^* \mid D) \approx \frac{1}{J} \sum_{j=1}^{J} p_M(D^* \mid \hat{w} + Pz_j), \quad z_j \sim p(z \mid D)$$

- Draw J samples from the posterior $p(z|D)$.
- Each sample z_j defines a new model $w_j = \hat{w} + Pz_j$.
- Average predictions $\rightarrow$ uncertainty-aware predictions.

How to Construct the Subspace?

Three main approaches:

1. **Random subspace:** Simple but uninformative
2. **PCA subspace:** From SGD trajectory
3. **Curve subspace:** Connects multiple solutions along low-loss curve

Random Subspace

- Sample K random vectors: $v_i \sim \mathcal{N}(0, I)$
- Normalize to unit length
- Use stochastic weight averaging solution as base: $\hat{w} = w_{SWA}$

Pros & Cons

- $+$ Virtually free to construct
- $+$ Simple baseline
- \- Poor model diversity
- \- Uncertainty doesn't grow away from data

PCA Subspace from SGD

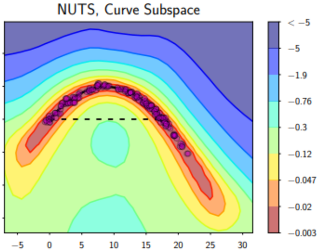
Construction Steps

1. Start from pretrained model
2. Run SGD with high LR
3. Collect weight snapshots w_i
4. Compute SWA: $\hat{w} = \text{mean}(w_i)$
5. PCA on deviations: $a_i = \hat{w} - w_i$
6. Take first K components as P

Why it works

SGD's oscillations reveal important directions. First PCA components align with top Hessian eigenvectors.

Curve Subspace



Key Idea

Connect two independent models with a low-loss curve in weight space.

Basis Construction

- Train two models: w_0 and w_1
- Find curve $w(t)$ with low loss between them
- $\hat{w} = (w_0 + w_1)/2$
- $v_1 = (w_0 - \hat{w})/\|w_0 - \hat{w}\|$
- $v_2 = (w_{1/2} - \hat{w})/\|w_{1/2} - \hat{w}\|$

Pros & Cons

- $+$ Excellent accuracy
- - Expensive ($3\times$ training cost)
- - Fixed 2D subspace

Bayesian Inference in the Subspace

Tempered posterior

To prevent over-concentration:

$$p_T(z|\mathcal{D}) \propto p(\mathcal{D}|z)^{1/T} p(z)$$

- $T = 1$: original posterior
- $T \to \infty$: approaches prior
- Chosen via cross-validation

Experiments: Regression (UCI Datasets)

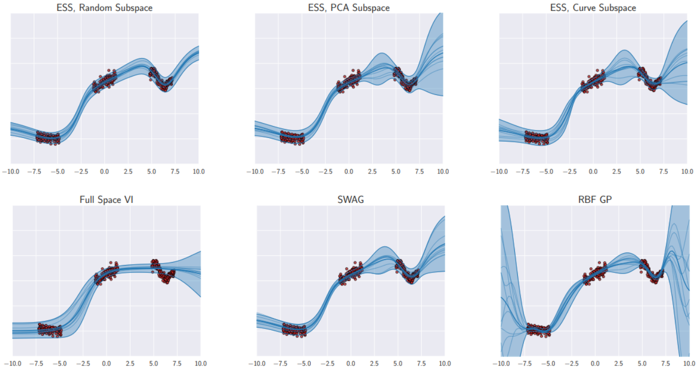


Figure: Predictive distributions for synthetic regression

Key observation: PCA and curve subspaces show uncertainty growing away from data.

Experiments: Image Classification (CIFAR)

Subspace	**NLL** $\downarrow$	**Accuracy** $\uparrow$
Random (10D)	0.6858 ± 0.0052	$80.17 \pm 0.03\%$
PCA (10D)	$\mathbf{0.6652 \pm 0.004}$	$\mathbf{80.54 \pm 0.13\%}$
Curve (2D)	$\mathbf{0.6493 \pm 0.01}$	$\mathbf{81.55 \pm 0.26\%}$

Table: Results for PreResNet-164 on CIFAR-100

- Even 2D curve subspace improves performance
- PCA subspace offers good trade-off between cost and performance

Conclusion

- Subspace Inference enables practical Bayesian deep learning
- Key insight: low-dimensional subspaces contain diverse, high-performing models
- PCA of SGD trajectory provides efficient, informative subspaces
- Method improves accuracy, calibration, and interpretability
- Bridge between SGD training and full Bayesian inference